Talk 3: KU and Adams operations

Priyanka

Draft v8: gray text is additional material.

1 Vector bundles and topological K-theory

Let $p\colon E \to B$ be a vector bundle.

Proposition 1.1. *Let B be compact Hausdorff. Then there exists E' such that $E' \oplus E \cong V^n \times B$ is a trivial vector bundle, where V^n denotes an n-dimensional vector space [6, §4, Thm. p. 13].*

Proposition 1.2. *Let E be a rank n vector bundle over a paracompact space B, with fibres over the field $K \in \{\mathbb{R}, \mathbb{C}\}$. Then there exists $b\colon B \to \mathrm{Gr}_n(K^\infty)$ such that $E \cong b^*E_n$, where E_n is the tautological bundle on the infinite Grassmannian. The map b is unique up to homotopy, so $[B, \mathrm{Gr}_n(K^\infty)] \cong \mathrm{Vect}^n_K(B)$ [4, Thm. 5.6, Thm. 5.7, §14].*

Definition 1.3. Let B be connected and compact Hausdorff. Then

$$K(B) = K(\mathrm{Vect}_{\mathbb{C}}(B), \oplus),$$

the Grothendieck group of the abelian monoid of isomorphism classes of complex vector bundles on B (cf. [5, Rem. 4.9]). The tensor product makes it a ring [6, §6, p. 16].

Remark 1.4. Any class can be written as a difference $[M] - [K^n]$, with M a vector bundle and $K^n = B \times K^n$ trivial: write $x = [E] - [F]$, choose G with $F \oplus G \cong K^n$ by Proposition 1.1, and then $x = [E \oplus G] - [K^n]$ [6, §6, p. 16f.].

Definition 1.5. $[E]$ and $[F]$ are said to be equivalent if

$$E \oplus V^n \cong F \oplus V^m$$

for some n, m ("stably equivalent"). Put

$$\widetilde{K}(B) = \mathrm{Vect}_{\mathbb{C}}(B)/\sim \cong \ker\big(K(B) \to K(*) \cong \mathbb{Z}\big),$$

where the map is restriction to a point of B, and the isomorphism sends $[E]$ to $[E] - [\mathbb{C}^{\mathrm{rk}\,E}]$ [6, §9, p. 32].

Definition 1.6. Negative K-groups:

$$\widetilde{K}^{-n}(X) = \widetilde{K}(\Sigma^n X),$$

where $\Sigma^n X$ is the n-fold reduced suspension [6, §10, p. 37].

Theorem 1.7. *K-theory and reduced K-theory are generalised (reduced) cohomology theories on compact (pointed) spaces [6, §9, Thm. p. 34; §10].*

Proposition 1.8. *$\widetilde{K}(B) = [B, BU]$, for B connected and compact Hausdorff (or a connected finite CW complex), where $BU = \operatorname{colim}_n BU(n)$ [2, Cor. III.15].*

Theorem 1.9 (Bott periodicity)**.**

$$\beta\colon \widetilde{K}(X) \xrightarrow{\sim} \widetilde{K}(\Sigma^2 X),$$

given by external multiplication with $\beta = [H] - 1 \in \widetilde{K}(S^2)$, where H is the hyperplane bundle on $\mathbb{C}P^1 = S^2$ [6, §9, Application (2), p. 34].

Remark 1.10 (Bott periodicity: the hyperplane bundle and the multiplication)*.*

1. $\mathbb{C}P^1 = \mathbb{P}(\mathbb{C}^2)$ is the space of lines $\ell \subseteq \mathbb{C}^2$, and $\mathbb{C}P^1 \cong S^2$. The tautological bundle $T \subseteq \mathbb{C}P^1 \times \mathbb{C}^2$ is the line bundle with fibre ℓ over the point ℓ. The *hyperplane bundle* is its dual $H = T^\vee$ [6, §5, Def. and Note (5), p. 14]. Since $\operatorname{rk} H = 1$, Definition 1.5 identifies $[H] - 1 \in \ker(K(S^2) \to K(*)) = \widetilde{K}(S^2)$, and this class generates $\widetilde{K}(S^2) \cong \mathbb{Z}$ [6, §9, p. 34].

2. For vector bundles $E \to X$ and $F \to Y$ the external tensor product $E \otimes F = \mathrm{pr}_X^* E \otimes \mathrm{pr}_Y^* F \to X \times Y$ has fibre $E_x \otimes F_y$ over (x, y). It is additive in E and F, so it induces a map of abelian groups $K(X) \otimes K(Y) \to K(X \times Y)$, $x \otimes y \mapsto x \times y$, and for pointed spaces it restricts to $\widetilde{K}(X) \otimes \widetilde{K}(Y) \to \widetilde{K}(X \wedge Y)$ [6, p. 38]. For $Y = S^2$ we fix the second factor to be $\beta \in \widetilde{K}(S^2)$ and get
$$\widetilde{K}(X) \longrightarrow \widetilde{K}(X \wedge S^2) = \widetilde{K}(\Sigma^2 X), \qquad x \longmapsto x \times \beta,$$
which is the map of Theorem 1.9.

2 Topological K-theory via anima

Notation 2.1.

$$\Pi_\infty\colon \mathrm{Top} \longrightarrow \mathrm{An}, \qquad \Pi_\infty\colon \mathrm{Grp}(\mathrm{Top}) \longrightarrow \mathrm{Grp}(\mathrm{An})$$

Here $\Pi_\infty X$ is the *fundamental ∞-groupoid* of a space X: the singular simplicial set $\operatorname{Sing} X$, which is a Kan complex [2, Ex. I.6(a)], regarded as an anima.

Definition 2.2. For a topological group G we define its classifying anima as

$$\widetilde{B}G := B\,\Pi_\infty(G) \in \mathrm{An},$$

where on the right $B\colon \mathrm{Grp}(\mathrm{An}) \to \mathrm{An}_*$ is the delooping of [5, Def. 3.3, Thm. 3.4]. The tilde distinguishes this anima from the topological classifying space BG (Remark 2.4), which is a space; Proposition 2.3 shows that the two agree after applying Π_∞.

Proposition 2.3. *Let $U = \operatorname{colim}_n U(n)$, the colimit taken in* Top*. There is an equivalence $\Pi_\infty BU \simeq \widetilde{B}U$ [2, Lemma* II.14c].*

Remark 2.4. For a topological group G, a *topological classifying space* is a space BG with a principal G-bundle $EG \to BG$ whose total space EG is contractible. It classifies principal G-bundles: for X paracompact, $[X, BG]$ is the set of isomorphism classes of principal G-bundles on X. For example $BU(n) = \mathrm{Gr}_n(\mathbb{C}^\infty)$ as in Proposition 1.2 (principal $U(n)$-bundles are rank n complex vector bundles with a hermitian metric), and $BU = \operatorname{colim}_n BU(n)$ is a classifying space for U. More generally $\Pi_\infty BG \simeq \widetilde{B}G$ for every topological group G with the homotopy type of a CW complex [2, Lemma* II.14c].

Now let X be a compact Hausdorff space.

Definition 2.5. Let $\mathrm{Vect}^{disc}(X)$ be the groupoid of complex vector bundles on X and their isomorphisms, regarded as an E_∞-monoid under $\oplus$ as in [5, Ex. 4.10]. Put

$$\mathrm{Vect}^{\simeq}(X) := \mathrm{colim}_{[n]\in\Delta^{op}} \mathrm{Vect}^{disc}\big(X \times |\Delta^n|\big).$$

For $X = *$ this is $\mathrm{Vect}_{\mathbb{C}}$ from [5, Ex. 4.12].

Proposition 2.6.

$$\mathrm{Vect}^{\simeq}(\mathrm{pt}) \simeq \coprod_{n\geq 0} \widetilde{B}U(n)$$

[5, Thm. 4.13], [2, Lemma II.14b].*

Definition 2.7.

$$\mathrm{BUP} := \mathrm{Vect}^{\simeq}(\mathrm{pt})^{gp} \simeq \mathbb{Z} \times \widetilde{B}U$$

(group completion as in [5, Thm. 4.7]) [2, Cor. III.15], [3, Ex. 3.2.23].

Remark 2.8 (comment by Maxime). Let $M = \mathrm{Vect}^{\simeq}(\mathrm{pt})$ and $s = [\mathbb{C}] \in \pi_0 M = \mathbb{N}$. The telescope is

$$T(M,s) = \mathrm{colim}\big(M \xrightarrow{\oplus\mathbb{C}} M \xrightarrow{\oplus\mathbb{C}} M \to \cdots\big) \simeq \mathbb{Z} \times \mathrm{colim}_n \widetilde{B}U(n) \simeq \mathbb{Z} \times \widetilde{B}U,$$

and there is a canonical map $T(M,s) \to M^{gp}$. By the group completion theorem [2, Thm. III.6], [3, Thm. 3.1.14] it induces an isomorphism on homology, $H_*(M^{gp}) \cong H_*(M)[\pi_0 M^{-1}] = H_*(T(M,s))$. Both sides have $\pi_0 = \mathbb{Z}$, and each component of $T(M,s)$ is simply connected, as $\pi_1(\widetilde{B}U) = \pi_0(U) = 0$. Each component of M^{gp} is simply connected as well: M^{gp} is an E_∞-group [5, Thm. 4.7, Constr. 4.8], so π_1 of a component is abelian and hence equal to $H_1 \cong H_1(\widetilde{B}U) = 0$. By Whitehead's theorem (a homology isomorphism between simply connected spaces is an equivalence) $T(M,s) \simeq M^{gp}$, i.e. $\mathrm{BUP} \simeq \mathbb{Z} \times \widetilde{B}U$ [2, Prop. III.12, Cor. III.15], [3, Prop. 3.2.10, Ex. 3.2.23]. For $M = \mathrm{Proj}_R$ with R a discrete ring the telescope is $K_0(R) \times BGL_\infty(R)$, and $\pi_1 = GL_\infty(R)$ is very non-abelian, so the same map is only a homology isomorphism. Recall $K(R) = \mathrm{Proj}_R^{gp}$ from [5, Def. 4.11]. One needs the plus construction: $K(R) \simeq K_0(R) \times BGL_\infty(R)^+$ [2, Prop. III.14, Cor. III.14a], [3, Ex. 3.2.22].

Remark 2.9.

$$\mathrm{Sp}_{\geq 0} \simeq \mathrm{CGrp}(\mathrm{An})$$

[2, Cor. II.24]. Under this equivalence (cf. [5, Not. 4.6] for CGrp), ku is the spectrum corresponding to BUP, so $\Omega^\infty \mathrm{ku} \simeq \mathrm{BUP} \simeq \mathbb{Z} \times \widetilde{B}U$.

Corollary 2.10. *Let X be a compact Hausdorff space with the homotopy type of a CW complex, e.g. a finite CW complex (see Remark 2.11). Then*

$$K^0(X) \cong \mathrm{ku}^0(\Pi_\infty X), \qquad \mathrm{ku}^0(\Pi_\infty X) \cong \pi_0 \mathrm{Map}_{\mathrm{Sp}}\big(\mathbb{S}[\Pi_\infty X], \mathrm{ku}\big),$$

where $\mathbb{S}[Y] = \Sigma^\infty_+ Y$ is the suspension spectrum.

Proof.

$$\begin{aligned} K^0(X) &\cong [X, \mathbb{Z} \times BU] \\ &\cong \pi_0 \mathrm{Map}_{\mathrm{An}}\big(\Pi_\infty X, \Pi_\infty(\mathbb{Z} \times BU)\big) \\ &\cong \pi_0 \mathrm{Map}_{\mathrm{An}}\big(\Pi_\infty X, \mathbb{Z} \times \widetilde{B}U\big) \\ &\cong \pi_0 \mathrm{Map}_{\mathrm{An}}\big(\Pi_\infty X, \mathrm{BUP}\big) \\ &\cong \pi_0 \mathrm{Map}_{\mathrm{Sp}}\big(\mathbb{S}[\Pi_\infty X], \mathrm{ku}\big). \end{aligned}$$

The steps are: Proposition 1.8 (plus the rank); Π_∞ being fully faithful on CW complexes up to homotopy [2, Cor. I.20]; Proposition 2.3; Definition 2.7; and the adjunction $\mathbb{S}[-] \dashv \Omega^\infty$ together with $\Omega^\infty \mathrm{ku} \simeq \mathrm{BUP}$. □

Remark 2.11. Without the CW hypothesis the second step of the proof fails. For example, the suspension of the Warsaw circle W is weakly contractible, so $\Pi_\infty \Sigma W \simeq *$, but $\widetilde{K}^0(\Sigma W) \neq 0$: we have $\widetilde{K}^0(\Sigma W) = K^{-1}(W) = [W, U] \supseteq [W, S^1] = \check{H}^1(W; \mathbb{Z}) = \mathbb{Z}$. Compare [2, Cor. III.15], which assumes X finitely dominated.

Proposition 2.12. *Let $\beta \in \pi_2(\mathrm{ku}) = \widetilde{K}(S^2)$ be the class of Theorem 1.9. Multiplication by β induces isomorphisms $\pi_n(\mathrm{ku}) \xrightarrow{\sim} \pi_{n+2}(\mathrm{ku})$ for all $n \geq 0$ [2, Thm. III.16].*

Remark 2.13. For $n \geq 1$ we have $\pi_n(\mathrm{ku}) = \pi_n(\Omega^\infty \mathrm{ku}) = \pi_n(\mathbb{Z} \times \widetilde{BU}) = [S^n, BU]_* = [S^n, BU] = \widetilde{K}(S^n)$, by Remark 2.9, Proposition 2.3 and Proposition 1.8 (unpointed and pointed maps agree, as BU is simply connected). Here $\widetilde{K}(S^n)$ is the reduced K^0 of the space S^n from Definition 1.5. Here the multiplication is the ring structure of ku: the tensor product of bundles distributes over $\oplus$, so the group completion ku of $(\mathrm{Vect}_{\mathbb{C}}, \oplus)$ is an E_∞-ring spectrum [2, III.15b], [3, Constr. 3.1.5]. Thus $\beta\colon \mathbb{S}^2 \to \mathrm{ku}$ gives $\Sigma^2 \mathrm{ku} = \mathbb{S}^2 \otimes \mathrm{ku} \xrightarrow{\beta \otimes \mathrm{id}} \mathrm{ku} \otimes \mathrm{ku} \to \mathrm{ku}$, which on homotopy groups is $x \mapsto \beta x$. Under $\pi_n(\mathrm{ku}) = \widetilde{K}(S^n)$ for $n \geq 1$ the product is the external product of Remark 1.10, so this is Theorem 1.9 for $X = S^n$. (In [2, III.16] the Hopf bundle T is used, so their β is $-([H] - 1)$.)

Corollary 2.14. *There are isomorphisms of graded rings*

$$\pi_* \mathrm{ku} \cong \mathbb{Z}[\beta] \quad \textit{and} \quad \pi_* \mathrm{KU} \cong \mathbb{Z}[\beta^{\pm 1}], \qquad |\beta| = 2,$$

where

$$\mathrm{KU} = \mathrm{ku}[\beta^{-1}] = \operatorname{colim}\Big(\mathrm{ku} \xrightarrow{\beta} \mathrm{ku}[-2] \xrightarrow{\beta[-2]} \mathrm{ku}[-4] \to \cdots\Big),$$

with $\mathrm{ku}[-2k] = \Sigma^{-2k}\mathrm{ku}$. *Since $\pi_n(\mathrm{ku}) = 0$ for $n < 0$, $\mathrm{ku} \simeq \tau_{\geq 0}\mathrm{KU}$ is the connective cover [2, after Thm. III.16], [3, Rec. 5.8.6].*

3 Adams operations

Definition 3.1 (λ-ring)**.** A *(pre-)λ-ring* is a commutative ring R with maps of sets $\lambda^i\colon R \to R$, $i \geq 0$, such that $\lambda^0(x) = 1$, $\lambda^1(x) = x$ and

$$\lambda^n(x + y) = \sum_{i+j=n} \lambda^i(x)\, \lambda^j(y),$$

i.e. $\lambda_t(x) = \sum_i \lambda^i(x) t^i$ is a homomorphism $(R, +) \to (1 + tR[[t]], \cdot)$.

A *λ-ring* additionally satisfies

$$\lambda^n(xy) = P_n\big(\lambda^1(x), \ldots, \lambda^n(x), \lambda^1(y), \ldots, \lambda^n(y)\big), \qquad \lambda^m(\lambda^n(x)) = P_{m,n}\big(\lambda^1(x), \ldots, \lambda^{mn}(x)\big),$$

where P_n and $P_{m,n}$ are certain universal integral polynomials. For their computation see [1, §I.1].

Definition 3.2 (Tensor power)**.** For a vector bundle $E \to X$ and $k \geq 0$, the k-th tensor power $E^{\otimes k}$ is the vector bundle with fibres $(E_x)^{\otimes k}$, obtained by applying the functor $V \mapsto V^{\otimes k}$ fibrewise [6, §3, p. 7].

Remark 3.3. If L is a line bundle, then $L^{\otimes k}$ is again a line bundle, since $\dim(L_x)^{\otimes k} = 1$.

Definition 3.4 (Exterior power)**.** For a vector space V the *k-th exterior power* is

$$\Lambda^k V = V^{\otimes k} \big/ \big\langle v_1 \otimes \cdots \otimes v_k \ : \ v_i = v_j \text{ for some } i \neq j \big\rangle,$$

Likewise $\Lambda^k E$ has fibres $\Lambda^k(E_x)$ [6, §3, p. 7]. One has $\Lambda^0 E = \mathbb{C}$, $\Lambda^1 E = E$, $\Lambda^k E = 0$ for $k > \operatorname{rk} E$, and

$$\Lambda^n(V \oplus W) \cong \bigoplus_{i+j=n} \Lambda^i V \otimes \Lambda^j W.$$

Theorem 3.5 (Splitting principle)**.** *For each vector bundle $\pi\colon E \to B$ over a compact Hausdorff space B there is a space $F(E)$ and a map $p\colon F(E) \to B$ such that $p^*E \to F(E)$ splits as a direct sum of line bundles. And*

$$p^*\colon K^*(B) \longrightarrow K^*(F(E))$$

is injective [6, §10, Splitting Principle, p. 41].

Proof. Let $\pi\colon E \to B$ have rank n and let $\mathbb{P}(\pi)\colon \mathbb{P}(E) \to B$ be the projective bundle, whose fibre over b is the projective space $\mathbb{P}(E_b)$ of lines in E_b. On $\mathbb{P}(E)$ there is the tautological line bundle $T \subseteq \mathbb{P}(\pi)^*E$, whose fibre over a line $\ell \subseteq E_b$ is ℓ itself. Choose a hermitian metric on E (possible as B is compact [6, §4, p. 9]).

Then $\mathbb{P}(\pi)^*E \cong T \oplus T^\perp$ with $T^\perp$ of rank $n-1$ [6, §3, p. 8]. Apply the same construction to $T^\perp \to \mathbb{P}(E)$, and so on. After $n-1$ steps we get the flag bundle $p\colon F(E) \to B$, an iterated projective bundle, whose fibre over b is the space of complete flags in E_b, and $p^*E \cong L_1 \oplus \cdots \oplus L_n$ with the L_i the pulled-back tautological line bundles [6, §5, Def. and Note (7), p. 15].

Injectivity. By the projective bundle theorem [6, §10, Prop. p. 41], $K^*(\mathbb{P}(E))$ is a free $K^*(B)$-module with basis $1, h, \ldots, h^{n-1}$, where h is the class of the hyperplane bundle $T^\vee$ (Remark 1.10). In particular $\mathbb{P}(\pi)^*$, which sends x to $x \cdot 1$, is injective. As p is a composite of projective bundle projections, p^* is a composite of injections, hence injective. □

Remark 3.6 (sketch of the projective bundle theorem). For a trivial bundle, $K^*(B \times \mathbb{C}P^{n-1}) \cong K^*(B)[h]/(h-1)^n$ follows from the periodicity theorem [6, §6, p. 17]. In general E is locally trivial, and one glues the local statements with Mayer–Vietoris sequences and the five lemma (a Leray–Hirsch argument) [6, §10, Thm. p. 41]. In fact $K^*(\mathbb{P}(E)) \cong K^*(B)[h]/\big(\sum_i (-1)^i [\Lambda^i E] h^i\big)$ [6, §10, Thm. p. 42].

Construction 3.7 (Adams operations $\psi^k\colon \mathrm{ku} \to \mathrm{ku}$)**.** 1. The exterior powers give

$$\lambda^i\colon \pi_0 \mathrm{Vect}_{\mathbb{C}}(X) \longrightarrow \pi_0 \mathrm{Vect}_{\mathbb{C}}(X), \qquad [V] \longmapsto [\Lambda^i V].$$

These are not additive, so we combine them into

$$\Lambda\colon \pi_0 \mathrm{Vect}_{\mathbb{C}}(X) \longrightarrow \mathrm{ku}^0(X)[[t]], \qquad [V] \longmapsto \lambda_t(V) = \sum_{i \geq 0} [\Lambda^i V]\, t^i.$$

By Definition 3.4,

$$\lambda_t(V \oplus W) = \lambda_t(V) \cdot \lambda_t(W).$$

Λ is a homomorphism into the multiplicative group $1 + t\,\mathrm{ku}^0(X)[[t]]$. Hence, by the universal property of group completion, it extends to $K^0(X) = \pi_0 \mathrm{Vect}_{\mathbb{C}}(X)^{gp}$ (for X as in Corollary 2.10; see [5, Thm. 4.7, Rem. 4.9]):

$$\lambda_t(V - W) = \frac{\lambda_t(V)}{\lambda_t(W)}.$$

Its coefficients give operations $\lambda^i\colon \mathrm{ku}^0(X) \to \mathrm{ku}^0(X)$ making $\mathrm{ku}^0(X)$ a λ-ring [6, §11, Ex. (3), p. 44f.].

By definition, $\mathrm{ku}^0(X) = \pi_0 \operatorname{Map}_{\mathrm{An}}(X, \mathbb{Z} \times \widetilde{B}U)$. So by the Yoneda lemma (in the homotopy category of An), natural operations on $\mathrm{ku}^0(-)$ are the same as homotopy classes of maps of anima $\mathbb{Z} \times \widetilde{B}U \to \mathbb{Z} \times \widetilde{B}U$. To extend the λ^i from finite CW complexes to all anima, one approximates $\mathbb{Z} \times \widetilde{B}U$ by Grassmannians [2, III.17, step (1)]. This yields maps of anima $\lambda^i\colon \Omega^\infty\mathrm{ku} \to \Omega^\infty\mathrm{ku}$, not respecting any E_∞-structure.

Remark 3.8 (from λ^i to ψ^k). The λ^i are natural, but not additive. The idea is to look at line bundles first. If $x = [L_1] + \cdots + [L_n]$ is a sum of line bundles, then $\lambda_t(x) = \prod_i (1 + [L_i]t)$, so $\lambda^k(x) = e_k([L_1], \ldots, [L_n])$ is the k-th elementary symmetric polynomial. The power sums $p_k = [L_1]^k + \cdots + [L_n]^k$ are additive in x, so we want $\psi^k(x) = p_k$. By Newton's identities the p_k are polynomials in the e_k, e.g. $p_1 = e_1$, $p_2 = e_1^2 - 2e_2$, and so ψ^k can be written as a polynomial in $\lambda^1, \ldots, \lambda^k$, which makes sense for every x. The generating function packages this: log turns the product $\lambda_{-t}(x) = \prod_i (1 - [L_i]t)$ into a sum, and

$$-t\,\frac{d}{dt} \log \lambda_{-t}(x) = \sum_i \frac{[L_i]t}{1 - [L_i]t} = \sum_{k \geq 1} p_k\, t^k.$$

This is the formula of the following definition. By the splitting principle (Theorem 3.5) the general case reduces to sums of line bundles.

Definition 3.9. For $x \in \mathrm{ku}^0(X)$ define $\psi^k(x)$ by

$$\sum_{k \geq 0} \psi^k(x)\, t^k = \mathrm{rk}(x) - t\,\frac{d}{dt} \log \lambda_{-t}(x) = \mathrm{rk}(x) - t\,\frac{\lambda'_{-t}(x)}{\lambda_{-t}(x)} \in \mathrm{ku}^0(X)[[t]].$$

The operation ψ^k is the *k-th Adams operation*. As λ_{-t} turns sums into products, $\log \lambda_{-t}$ and hence every ψ^k is additive. Since the λ^i are natural in X, so is ψ^k, and by the Yoneda argument of Construction 3.7 it is a map of anima $\psi^k\colon \mathbb{Z} \times \widetilde{B}U \to \mathbb{Z} \times \widetilde{B}U$, i.e. $\psi^k\colon \Omega^\infty\mathrm{ku} \to \Omega^\infty\mathrm{ku}$.

Proposition 3.10. *For all $x, y \in \mathrm{ku}^0(X)$ and $k, l \geq 0$:*

1. *$\psi^k[L] = [L]^k = [L^{\otimes k}]$ for a line bundle L;*
2. *ψ^k is a ring homomorphism;*
3. *$\psi^k \circ \psi^l = \psi^{kl}$;*
4. *$\psi^p(x) \equiv x^p \bmod p$ for p prime;*
5. *$\psi^k(\beta^n) = k^n \beta^n$ for $\beta^n \in \widetilde{K}(S^{2n}) = \pi_{2n}(\mathrm{ku})$.*

Moreover, ψ^k is the unique natural additive operation satisfying (1) [6, §11, Prop. and Remarks, p. 45], [2, Prop. III.18].

Sketch. (1): $\lambda_{-t}[L] = 1 - [L]t$, so $-t\frac{d}{dt}\log(1 - [L]t) = \sum_{k \geq 1} [L]^k t^k$. For (2)–(4) and uniqueness, by naturality and the splitting principle (Theorem 3.5) we may pull back to $F(E)$ and assume that all bundles involved are sums of line bundles; there all claims follow from additivity and (1), e.g. $\psi^p(\sum_i L_i) = \sum_i L_i^p \equiv (\sum_i L_i)^p \bmod p$. For (5): $\psi^k(\beta) = [H]^k - 1 = \beta \sum_{j<k} [H]^j = k\beta$, as $\beta^2 = 0$ in $\widetilde{K}(S^2)$, and then use multiplicativity. □

References

[1] M. F. Atiyah, D. O. Tall, *Group representations, λ-rings and the J-homomorphism*, Topology **8** (1969), 253–297.

[2] F. Hebestreit, F. Wagner, *Algebraic and Hermitian K-theory*, lecture notes, Bonn. https://florianadler.github.io/AlgebraBonn/KTheory.pdf.

[3] K. Hilman, J. McCandless, *Lecture notes on algebraic K-theory*, MPIM Bonn, 2024. https://algebraicktheorywise26-27-66b76f.gitlab.io/Hilman-McCandless-latest.pdf.

[4] J. Milnor, J. Stasheff, *Characteristic Classes*, Annals of Math. Studies **76**, Princeton University Press, 1974.

[5] Talk 1 of this seminar, *Semiadditivity, additivity, stability* (Zauri), notes.

[6] K. Wirthmüller, *Vector bundles and K-theory*, skeleton lecture notes, Kaiserslautern, 2011/12. https://ncatlab.org/nlab/files/wirthmueller-vector-bundles-and-k-theory.pdf.